The Intelligence Cost of Surveillance*

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Abstract

Surveillance deters dissent, but the deterrence is purchased with the regime's own intelligence. In a regime-change coordination game with pre-play communication, citizens who believe they are monitored still write messages—volumes barely change—but in a guarded register that destroys information selectively: a guarded "nothing is happening" reads like the direct version; a guarded "I will be in the square tonight" does not. In a preregistered held-out test, every analyst model reading surveilled messages becomes significantly worse at judging regime survival precisely in weak-regime states (all $p < 0.0001$; strong-state placebos clean), a frontier-capability analyst loses as much as models costing one-fiftieth as much, and a supervised classifier replicates the loss. Both the chilling and the blinding arrive at the mildest visible monitoring cue. Feeding the measured curves into the regime's problem: if the regime can defeat more than a quarter of fully anticipated uprisings, visible surveillance *lowers* its survival. Surveillance suppresses the mean of unrest and fattens its tail.

JEL: C72, C92, D82, D83, P16
Keywords: surveillance, self-censorship, coordination, regime change, authoritarian intelligence, LLM agents

1 Introduction

New communication channels make dissent mutually visible. Cassette tapes carried Khomeini's sermons into Iran in 1979; fax machines threaded through Eastern Europe in 1989; Facebook and Twitter synchronized the squares of 2011; Telegram channels steered Minsk and Hong Kong in 2019–2020; and in September 2025, Nepali protesters organized on Discord—a platform their government had not thought to watch—within days of a social-media ban.[1] In each case the channel's contribution was not news—grievances were old—but *legibility*: each citizen could finally read that others were willing to act.

Regimes respond by monitoring the channel, and the standard account stops there: monitoring chills speech, coordination fails, order returns. This paper supplies the account's missing half. Monitored citizens do not stop talking; they change *how* they talk, substituting from direct,

*Draft: June 2026. Companion to "Speaking in Code: How Surveillance Suppresses Coordinated Dissent." All experiments preregistered; the full preregistration history, including one failed gate, is reported in Appendix B. Replication code and data are in the public repository. I thank [advisor, seminar participants] for comments.

[1] Reuters, September 9, 2025, "Nepal lifts social media ban after protests leave 19 dead." The ban itself helped trigger the protests; press reports indicate Discord deliberation figured in the selection of the interim prime minister.

incriminating, coordination-relevant language into a guarded, deniable register. That substitution is what suppresses coordination—the subject of the companion paper [Eltokhy, 2026]—but the same substitution degrades the signal the regime extracts from listening. Surveillance is not a pure monitoring technology. It is simultaneously a data-destruction technology, and the destruction is aimed, by the logic of self-censorship itself, at exactly the information the monitor values most.

The central empirical finding is that this intelligence cost is *crisis-concentrated*. I embed large-language-model (LLM) citizens in a Morris and Shin [2003] regime-change game with a pre-play messaging stage, hold the state, the senders, and their private signals fixed, and vary only whether messages are written under a monitoring warning. "Regime analyst" models then read the intercepted messages and answer the questions an intelligence service asks: How likely is the regime to fall? What share of these citizens will join? Which ones? In strong-regime states, analysts read surveilled traffic as well as free traffic—occasionally better, since uniformly guarded calm is easy to classify. In weak-regime states, where uprisings actually succeed, every analyst's judgment of regime survival deteriorates sharply: across every analyst-model and corpus combination in the program, fourteen of fourteen tests, with the decisive test preregistered on held-out data and a single primary outcome fixed in advance (all four analysts, permutation $p < 0.0001$; all strong-state placebos clean). The deterioration peaks not in hopelessly weak states, which are easy to call even from guarded text, but in the ambiguous tipping-point band where the regime's response decisions carry their value. The regime's average intelligence metrics would never reveal the problem. The gauge reads normal until the boiler bursts.

Three results sharpen the mechanism, and a fourth prices it. First, the blindness is in the data, not the reader: a frontier-capability analyst—the best model on every level metric—loses as much in crisis states as models costing one-fiftieth as much, and a supervised text classifier trained on free speech reproduces the crisis-concentrated loss, so no reader, statistical or frontier-AI, can extract content that senders never encoded. Second, the text shows why: under a monitoring warning, essentially *all* senders shift into a guarded register (a classifier separates the arms at AUC 0.93–0.97 in every signal bin), but the shift destroys information only where incriminating information existed—compliance is universal, destruction is selected. Third, dose-response experiments across graded warnings show both margins are step functions of monitoring salience: the mildest cue, with no consequence language at all, already delivers two-thirds of the chilling and roughly four-fifths of the blinding; the regime's real choice is not how much to monitor but whether to be seen monitoring. Fourth, feeding the measured chilling and blinding curves into the regime's survival problem yields a quantitative bottom line: on the experimental state distribution, visible surveillance raises regime survival only if the regime can defeat fewer than 26% of fully anticipated uprisings; for any more capable responder—or any regime with higher crisis exposure—the intelligence cost exceeds the deterrence benefit.

The theoretical contribution is a two-sided model of visible surveillance in a global game with horizontal communication. Citizens' equilibrium response to monitoring is a common guarded register, formalized as *camouflage pooling*: messages expressing more regime weakness than an incrimination threshold are replaced by draws from an anodyne distribution indistinguishable from genuinely moderate reports. Two results follow with little machinery. Because the censored region is populated in proportion to the regime's weakness, the informativeness loss to any reader is concentrated in weak states—crisis blindness as a selection theorem, not an assumption—and because the same garbling degrades peers' reading of each other's willingness, one instrument prices both effects. A corollary formalizes why the cost is undetectable from inside: average intelligence quality can be arbitrarily close to its unmonitored level while crisis-state intelligence is destroyed. The model's comparative static matches the data's surprise: the trade-off binds at the extensive margin—monitor visibly or not at all—and visible monitoring is favored only by regimes whose response capacity is too weak for intelligence to matter.

This paper relates to four literatures. The political economy of authoritarian information control establishes that surveillance and censorship suppress dissent [King et al., 2013, Roberts, 2018, Xu, 2021, Guriev and Treisman, 2019]; the dictator's-dilemma tradition [Wintrobe, 1998, Egorov et al., 2009, Lorentzen, 2014, Gehlbach et al., 2026] argues repression degrades the autocrat's knowledge of society. The closest paper is Yang [2025], who shows that repression-induced silence corrupts the training data of AI censorship systems. The mechanism here is distinct and strategic: the monitored population does not fall silent—message volumes barely change—but re-encodes, so the regime's intelligence problem is produced in equilibrium by its own advertised monitoring, and the citizen-coordination side is priced within the same model. Second, the global-games literature provides the coordination framework [Carlsson and van Damme, 1993, Morris and Shin, 2003, Angeletos et al., 2007]; existing work on manipulated information in regime change is vertical—a designer chooses citizens' signals [Edmond, 2013, Inostroza and Pavan, 2025]—whereas here citizens talk to each other and the regime eavesdrops, a structure with essentially no incumbent model. Communication under an adversarial listener appears in industrial organization [Spector, 2022] and in Braghieri [2024] on political correctness; I port it to the regime-change setting and add the listener's inference problem as an outcome. Third, the methodology builds on LLM agents as experimental subjects [Horton, 2023, Mei et al., 2024], used here in the register the recent methodological literature licenses [Ludwig et al., 2026]: as a controlled measurement instrument for the informativeness of message distributions, validated by falsification structure—preregistration with a public failure on record, held-out confirmation, placebo states, model-free classifier replication—rather than by claimed human equivalence. Fourth, the mechanism connects to AI-safety work on monitorability: optimizing models against chain-of-thought monitors teaches obfuscated misbehavior [Baker et al., 2025, Motwani et al., 2024]. Autocrats and AI labs face the same theorem: monitoring pressure selects for less monitorable communication.

Section 2 presents the model. Section 3 describes the design and the preregistration structure. Section 4 reports the blinding results. Section 5 opens the text. Section 6 reports the dose-response and solves the regime's problem with measured inputs. Section 7 concludes. Appendices report prompts, warning texts, the full preregistration history, and additional results.

2 Model

2.1 Environment and timing

A continuum of citizens $i \in [0,1]$ and a regime. Regime strength is $\theta \in \mathbb{R}$; low θ is weak. Citizens receive private signals $x_i = \theta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d. An uprising succeeds if and only if the joining mass exceeds the threshold: $A \equiv \int a_i \, di > \theta$, with payoffs B to joining a successful uprising, $-C$ to joining a failed one, 0 to staying.

Timing:

t_0. The regime publicly commits to visible monitoring $m \in \{0,1\}$ before θ is realized. Only *believed* monitoring affects behavior; rational expectations equate advertised and believed monitoring.[2]

t_1. Nature draws θ and signals x_i.

t_2. Each citizen sends a message to peers; the regime intercepts the message vector M.

t_3. The regime reads M and forms a posterior $\hat{q}$ over the success event $\{A > \theta\}$.

t_4. Citizens choose $a_i \in \{0, 1\}$.

t_5. If $A > \theta$, the regime's response (preemption, targeted arrests, concession) defeats the uprising with probability $H(I_R)$, where I_R is the informativeness of intercepted traffic and $H' > 0$. Otherwise the regime falls.

Placing the response's *effect* at t_5, after actions, separates the channels cleanly: regime survival is

$$S(m) \;=\; 1 - \mathbb{E}_\theta\Big[\, U(m, \theta)\,\big(1 - H(I_R(m, \theta))\big) \Big], \tag{1}$$

where $U(m, \theta) = \Pr(A > \theta \mid m, \theta)$ is the probability the citizen side coordinates past the threshold.

2.2 Message technology: the guarded register

Absent monitoring, truthful communication ($\mu_i = x_i$) is an equilibrium of the messaging stage: citizens share the goal of coordinating on the correct action, so cheap talk is self-enforcing as in common-interest games.[3] Under monitoring, a message whose content reveals a signal weaker (more anti-regime) than an incrimination threshold $\bar{x}$ exposes the sender to punishment.

Assumption 1 (Incrimination)**.** *Under $m = 1$, a sender whose message reveals $x_i < \bar{x}$ pays $\kappa > 0$ in expectation. Messages revealing $x_i \geq \bar{x}$ carry no risk.*

Lemma 1 (Guarded register)**.** (a) (Compliance.) *If κ exceeds each type's private value of revealing x_i, all citizens adopt the guarded register under $m = 1$: the message map becomes*

$$c(x_i) \;=\; \begin{cases} x_i & x_i \geq \bar{x}, \\ \tilde{x} \sim \bar{F} & x_i < \bar{x}, \end{cases}$$

where $\bar{F}$ is an anodyne distribution supported on $[\bar{x}, \bar{x} + \delta]$—camouflage pooling: guarded messages from incriminated types are indistinguishable from genuinely moderate reports. (b) (Selected destruction.) *For every θ, the monitored message experiment is a Blackwell garbling of the unmonitored one, and for any decision problem with bounded payoffs the value loss from monitoring is at most proportional to the censored mass $\Phi\big((\bar{x} - \theta)/\sigma\big)$ per message.*

Proof. (a) For $x_i \geq \bar{x}$, c is the identity and compliance is free. For $x_i < \bar{x}$, defying the register risks κ, while complying costs the marginal contribution of one message to the sender's own

[2] Covert interception is costless and inframarginal: it changes no citizen's behavior and weakly raises the regime's information, so it is always undertaken. The economics is entirely in visibility. A signaling extension in which advertised and actual monitoring diverge is discussed in Section 7.

[3] See Avoyan [2024] on pre-play communication in global games. The experiment implements messages as free-form prose; the companion paper verifies that peer messages carry decision-relevant information.

coordination payoff—negligible in the continuum and $O(1/n)$ with n peers. For κ above this bound, compliance is dominant.

(b) Define the Markov kernel K on the message space: $K(\cdot \mid \mu) = \delta_\mu$ if $\mu \geq \bar{x}$ and $K(\cdot \mid \mu) = \bar{F}$ if $\mu < \bar{x}$. Because unmonitored messages equal signals, the monitored message distribution satisfies $P_\theta^1 = K \circ P_\theta^0$ for *every* θ, with K independent of θ: the monitored experiment is a garbling of the unmonitored experiment, hence Blackwell-dominated, which proves the weak inequality for any informativeness functional respecting the Blackwell order. For the localization, couple the experiments message by message: they produce identical output on the event $\{x_i \geq \bar{x}\}$, which has probability $1 - \Phi((\bar{x} - \theta)/\sigma)$. For any decision problem with payoff range R, the per-message value difference is bounded by $R \cdot \Phi((\bar{x} - \theta)/\sigma)$, the probability of the only event on which the experiments differ. □

Remark 1 (What the data demanded). An earlier version derived crisis-concentration from a coding *choice* cutoff: only incriminated types pay the cost of coding. The text data reject the choice margin's empirical bite and support Lemma 1's structure directly: under a warning, the *stylistic* shift is near-universal—an arm classifier separates monitored from free messages at AUC 0.93–0.97 in every sender-signal bin (Section 5)—while the *informational* loss is selected. Compliance is a trigger response to monitoring salience, consistent with Kuran [1991]'s discrete public persona, not a marginal-deterrence response; and Lemma 1 requires only that the guarded register pools the incriminated, which is weaker than a cutoff in the coding choice.

2.3 Two readers, one garbling

Citizens and the regime read the same transformed message vector. Let $I_R(m, \theta)$ denote a monotone informativeness functional of intercepted traffic about the objects the regime needs—the state and the attack mass—and $L_C(m, \theta)$ the corresponding informativeness to a receiving citizen about peers' willingness.[4] Define the legibility wedge $W = L_C - L_R$; coding is steganographic if $W > 0$. The experiments find $W \approx 0$: both readers lose. Equilibrium coding here is mutually destructive camouflage, not an in-group code.

2.4 Results

Proposition 1 (Two-price surveillance). *Under Assumption 1 and Lemma 1, switching* $m : 0 \to 1$ *(i) weakly lowers* $U(\cdot, \theta)$ *for all* θ *(chilling), and (ii) weakly lowers* $I_R(\cdot, \theta)$ *for all* θ *(blinding), strictly wherever* $\Pr(x_i < \bar{x} \mid \theta) > 0$.

Proof. (ii) is Lemma 1(b) applied to the regime's decision problem. (i) follows from the same garbling applied to the receiving citizen's problem: each citizen's posterior about peers' willingness under $m = 1$ is a garbling of the posterior under $m = 0$, so the informational content supporting coordination weakly falls; in the threshold equilibrium of the communication game the attack mass is weakly decreasing in the informativeness of peer messages about willingness on the relevant region. The companion paper verifies (i) directly: joining falls by 8.0 percentage points on the same grid ($t = -16.2$, $N = 500$ matched cells). □

Proposition 2 (Crisis blindness). *The intelligence loss* $I_R(0, \theta) - I_R(1, \theta)$ *is decreasing in* θ, *strictly positive wherever the censored region has mass, and vanishes as* $\theta \to \infty$.

[4] Any functional respecting the Blackwell order suffices; the experiment operationalizes the state margin as Brier accuracy of a regime-fall probability, the aggregate margin as join-share error, and the individual margin as sender-level classification.

Proof. Immediate from the localization in Lemma 1(b): the experiments coincide outside the censored event, whose probability $\Phi((\bar{x}-\theta)/\sigma)$ is strictly decreasing in θ with limit zero. □

Corollary 1 (The calibration trap)**.** *For any $\epsilon > 0$ there exist priors G over θ with $\mathbb{E}_G[I_R(0,\theta) - I_R(1,\theta)] < \epsilon$ while $I_R(0,\theta) - I_R(1,\theta)$ remains bounded away from zero on $\{\theta < 0\}$: a regime auditing its surveillance apparatus on average performance will not detect the crisis blindness.*

Proposition 3 (The extensive margin)**.** *From (1),*

$$S(1) - S(0) \;=\; \underbrace{\mathbb{E}_\theta\big[(U_0 - U_1)(1 - H(I_{R,1}))\big]}_{\textit{chilling gain } \geq 0} \;-\; \underbrace{\mathbb{E}_\theta\big[U_0\,(H(I_{R,0}) - H(I_{R,1}))\big]}_{\textit{blinding loss } \geq 0},$$

where subscripts denote the monitoring state. Visible monitoring lowers survival when response capacity is strong and crisis exposure high—H' large and prior mass on low θ large, where (Proposition 2) the blinding term concentrates and (because U_0 is largest there) is most heavily weighted—and raises it when intelligence cannot be acted on ($H' \approx 0$). Section 6 evaluates this expression with measured inputs.

Proposition 4 (Suppressed mean, fattened tail)**.** *Let $\theta \sim G$ across periods, let $\hat{q}(m,\theta) = \mathbb{E}[\mathbf{1}\{A > \theta\} \mid M_m]$ be the regime's t_3 posterior on the success event, and suppose the garbling of Lemma 1(b) shrinks posteriors toward the prior on the censored region. Then visible monitoring (i) lowers the unconditional probability of successful uprisings, and (ii) raises the regime's expected surprise conditional on a successful uprising, $\mathbb{E}[\,1 - \hat{q} \mid A > \theta\,]$.*

Proof. (i) is Proposition 1(i) integrated over G. For (ii): successful uprisings concentrate on the crisis region, where the censored mass is large; on that region the truth of the success event is 1 while the unconditional prior probability of success is strictly below 1, so shrinkage toward the prior strictly lowers $\hat{q}$ on the conditioning event, raising $1 - \hat{q}$ in expectation. The shrinkage premise follows from camouflage pooling: messages from the censored region are uninformative about θ beyond the (unidentified) censoring event, so the Bayesian posterior given heavily censored traffic collapses toward the prior. The measured counterpart: on cells where the uprising succeeds in crisis states, the pooled analyst posterior is 0.69 reading free traffic and 0.59 reading surveilled traffic (Section 6). □

3 Experimental design

The design exports the companion paper's language-based global game [Eltokhy, 2026] and adds the regime's side of the table. Citizens are LLM agents who receive natural-language intelligence briefings generated from private signals, exchange free-form messages with trusted contacts, and decide JOIN or STAY. The state grid is 10 countries × 50 periods (500 cells, $n = 25$ agents, seeded), with realized θ, every sender's message, and every sender's decision archived per cell. The surveillance treatment appends a monitoring warning to the message-writing prompt only (verbatim texts in Appendix A); the briefing, the decision prompt, and the decision environment are bit-identical across arms, and the arms share the same θ draws and per-agent signal vectors by construction.

The analyst task. For each cell and arm, a "regime analyst" model receives the cell's intercepted messages—shuffled, neutrally labeled, with no arm indicators, no state information, and never the senders' reasoning—and returns, in a strict parseable format: (i) the probability the regime falls within the month; (ii) the percentage of shown senders who chose JOIN; (iii) a JOIN/STAY verdict and join probability per sender (Appendix A). Ground truth for all three is in the archive. The roster spans a capability gradient: four 2026-generation open models (DeepSeek v4 flash, Llama 4 Maverick, Qwen 3.7 Plus, GLM 5.1), the 2025-generation Llama 3.3 70B, and a frontier model (Claude Opus 4.8). Temperature 0; analyst prompts identical across arms; the only arm-varying content is the message text, which is the treatment. An analyst inferring the arm from style is not a confound—guarded style is the equilibrium object the regime must read through.

Identification. Because θ, senders, and signals are held fixed within a cell pair, any difference in analyst accuracy across arms is caused by the surveillance-induced change in the messages. The program proceeded in three preregistered stages, reported in full in Appendix B: an exploratory pilot whose targeting metric proved mis-designed (v1); a confirmatory specification that formally failed (v2, NOT CONFIRMED)—the failure isolated to the v2 targeting metric and itself informative (Section 4.3); and a held-out confirmation (v3) with a single primary outcome—crisis-state Brier—on 348 cells never sampled by any prior run, with the strong-state cells as the model-implied placebo. Robustness uses sparse interception (10 of 25 messages), a second sender family (Qwen3-30B corpus, 246 matched cells), a supervised-classifier analyst, and a codedness-control corpus of equally indirect messages produced without monitoring framing.

4 Results: surveillance blinds the watcher in crisis

4.1 The crisis-state result and its held-out confirmation

Table 1 reports the program's central estimate—the crisis-state ($\theta < 0$) paired Brier delta on the regime-survival judgment—across every wave and analyst. The effect appears in fourteen of fourteen tests. The decisive evidence is the preregistered held-out wave: on 348 cells untouched by any prior run, with crisis Brier fixed in advance as the single primary, all four analysts degrade (deltas -0.047 to -0.121, permutation $p < 0.0001$ on 181 crisis cells) and all four strong-state placebos are clean.

Figure 1(a) shows the result's structure. The blinding is not a uniform haze: surveilled and free traffic are equally readable in strong states and in hopelessly weak ones, and the gap concentrates in the band where the outcome hangs on coordination—the precise region where preemption and concession decisions are made.

4.2 The blindness is in the data, not the reader

The frontier analyst is the best model on every level metric (e.g., $|\rho(\text{FALL}, \theta)| = 0.93$, the highest in the program) and loses as much in crisis as models costing two orders of magnitude less: crisis ΔBrier -0.085 ($p < 0.001$), crisis sender-accuracy loss 6.5 points ($p = 0.003$), with significant calm-state *reversals*—surveillance improves its calm-state reading, because uniformly guarded quiet is easy to classify (Figure 1(b)). Capability raises the level of intelligence everywhere; it does not protect the crisis margin, because no reader can extract content the senders never encoded.

Table 1: Crisis-state blinding across the full program: paired ΔBrier (baseline − surveillance), $\theta < 0$ cells.

Wave	Analyst	ΔBrier	p_{perm}	Placebo ($\theta \geq 0$) clean
Pilot, full interception (150 cells; exploratory)	DeepSeek v4 flash	−0.112	< 0.001	✓
	Llama 4 Maverick	−0.047	< 0.001	✓
	Qwen 3.7 Plus	−0.076	0.001	✓
	GLM 5.1	−0.137	< 0.001	✓
	Llama 3.3 70B	−0.052	< 0.001	✓
	Claude Opus 4.8	−0.085	< 0.001	(reversed)
Sparse interception, $k = 10$ (150 cells; confirmatory v2)	DeepSeek v4 flash	−0.100	< 0.001	✓
	Llama 4 Maverick	−0.059	< 0.001	✓
	Qwen 3.7 Plus	−0.099	< 0.001	✓
	Llama 3.3 70B	−0.080	< 0.001	✓
Held-out, preregistered v3 (348 fresh cells; single primary)	DeepSeek v4 flash	−0.100	< 0.0001	✓
	Llama 4 Maverick	−0.049	< 0.0001	✓
	Qwen 3.7 Plus	−0.121	< 0.0001	✓
	Llama 3.3 70B	−0.047	< 0.0001	✓

Notes: Δ = baseline − surveillance; negative = analysts more accurate on baseline messages. 10,000-draw sign-flip permutation tests, seeds fixed ex ante. "Placebo clean" = no significant blinding at $\theta \geq 0$; "reversed" = surveillance significantly *improves* calm-state accuracy (Opus 4.8: +0.022, $p = 0.012$), as camouflage pooling predicts when the censored region is empty. v3 verdict: CONFIRMED, 4/4 primaries, 4/4 placebos. Full preregistration history in Appendix B.

A supervised classifier (TF-IDF and logistic regression with cell-level cross-fitting) makes the result model-free: trained on free speech, its sender-level AUC falls from 0.922 to 0.844 on surveilled crisis traffic, double the calm-state gap. Granting the regime labeled surveilled data to retrain on—the best case, and precisely the data Yang [2025] shows regimes lack—recovers only part of the loss (crisis AUC 0.909).

Cross-sender robustness is, if anything, stronger: on a second sender family (Qwen3-30B corpus, 246 matched cells), pooled sender AUC falls 0.81 → 0.67 and 0.89 → 0.75 (both $p < 0.0001$), crisis sender-accuracy losses reach 17–18 points, and even the aggregate join-share margin—null on the primary corpus—degrades significantly. The largest effects in the program are cross-family readers on a non-Llama corpus, which retires the concern that same-family writer-reader recognition drives the result.

4.3 What survives: the decomposition of the loss

Two margins are robust to surveillance, and their robustness is informative. Aggregate temperature: analysts' join-share estimates barely move on the primary corpus (MAE 0.17–0.25 in both arms)—the regime can still tell that people are unhappy. Within-cell ranking: the per-cell ranking AUC of senders is statistically flat in nearly every test; this is the isolated reason the v2 confirmatory gate returned NOT CONFIRMED (its conjunctive primary required this metric to move; Appendix B). The loss concentrates in *between-cell, state-level* discrimination: which cells are dangerous, and whether the state is one where coordination will tip. Coding hides the fire, not the relative ordering of the people standing near it. This is the politically decisive margin:

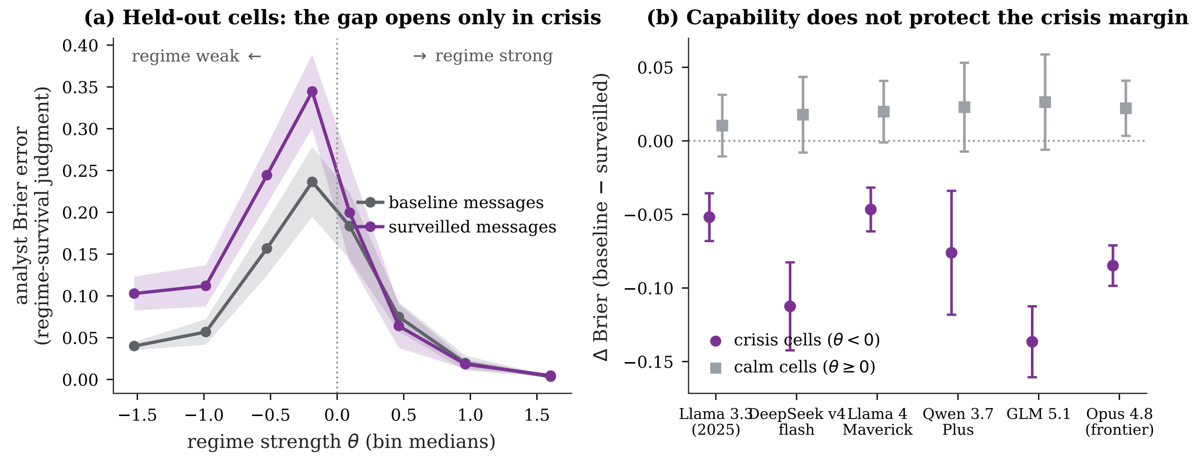

Figure 1: **Crisis blindness.** (a) Analyst Brier error on regime survival by state, pooled over the four held-out analysts (348 preregistered fresh cells; cell-level 95% bands). The arms coincide for $\theta \geq 0$ and diverge below it—and the gap peaks not in hopeless states, which are easy to call even from guarded text, but in the *ambiguous tipping-point band*, exactly where the regime's response decision has value. (b) Paired crisis (purple) and calm (gray) Brier deltas by analyst, ordered by capability tier, on the main 150-cell sample. The frontier model loses as much in crisis as models costing two orders of magnitude less.

preemption, concession, and timing run on tipping-point detection—the object $H(\cdot)$ acts on in (1)—and it is the margin Corollary 1 says the regime cannot audit from averages.

5 Mechanism: the text

Universal compliance, selected destruction. Across 24,306 matched messages, a classifier on simple text features separates surveilled from free messages at AUC 0.93–0.97 in *every* sender-signal and state bin: essentially everyone shifts register under a warning (Figure 2). The divergence gradient is nonetheless monotone in the predicted direction (Spearman $\rho = -0.79$, $p = 0.02$ across sender-signal bins; $\rho = -0.98$, $p < 0.001$ across θ bins), but its magnitude makes clear that the style response is a trigger, not a margin. The information loss, by contrast, is sharply selected (Section 4): a guarded version of "nothing is happening" has the same likelihood ratio as its direct counterpart; a guarded version of "people I trust are ready to move" does not. This is Lemma 1 read off the data, and it is why the model needs no coding-choice cutoff.

Style alone is not enough. On a codedness-control corpus—equally indirect, metaphorical messages elicited *without* monitoring framing—analyst θ-recovery degrades directionally for all three analysts tested but reaches at most marginal significance ($|\rho|$: 0.35 vs. 0.21, $p = 0.06$; 0.44 vs. 0.38; 0.42 vs. 0.37). Combined with the companion paper's finding that coded style alone does not move *receivers'* actions, the blinding travels with what monitored senders choose not to convey, more than with indirection as a form.

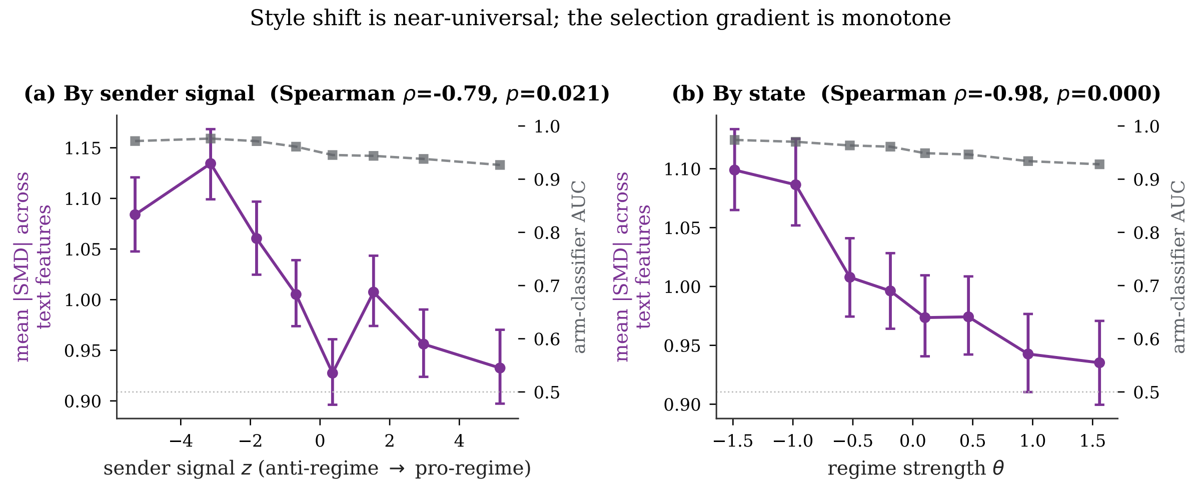

Figure 2: **Universal compliance, selected destruction** (24,306 matched messages). Purple: mean absolute standardized difference of text features (action cues, direct regime terms, hedges, commitment language, length) between arms, by sender-signal bin (a) and state bin (b), with bootstrap 95% CIs. Gray (right axes): AUC of a classifier separating the arms on the same features. The style shift is near-universal—the arms are separable at AUC 0.93–0.97 in every bin—while the divergence gradient is monotone in the predicted direction. The information loss (Figure 1) is far more selected than the style shift: Lemma 1's structure, read off the text.

6 Dose-response and the regime's problem

6.1 Both margins arrive at the mildest cue

Graded warnings on the identical 500-cell grid trace both curves (Figure 3). Chilling (mean join fraction): none 0.471, mild 0.408, full 0.391, severe 0.376—monotone, strongly concave, two-thirds of the total at the mildest cue, which contains no consequence language at all. Blinding (crisis ΔBrier, three analysts): the mild dose already delivers 75–85% of the full-dose effect (e.g., DeepSeek -0.082 mild, -0.112 full, -0.110 severe; all $p < 0.0001$ at every dose). Both effects arrive almost entirely at the extensive margin, supporting the trigger interpretation of compliance (Remark 1) and the binary statement of the regime's problem (Proposition 3). Conditional on visible monitoring, escalating severity buys additional silence (-6.3 to -9.5 pp) at little additional blindness—within the monitoring regime, severity is roughly free. The costly decision is being seen to monitor at all.

6.2 The regime's problem with measured inputs

Equation (1) can now be evaluated with every input measured (Figure 4). For each of the 498 matched cells and each arm, the citizen side supplies the realized uprising outcome (chilling lowers success: 0.582 baseline vs. 0.566 surveilled on identical states), and the regime side supplies the detection posterior $\hat{q}$—the pooled FALL probability of the four common analysts reading that cell's traffic. On the success events that kill regimes, detection is what monitoring destroys: in crisis cells where the uprising succeeds, $\hat{q}$ falls from 0.69 (free traffic) to 0.59 (surveilled traffic), the measured counterpart of Proposition 4(ii). Closing the model with the linear response

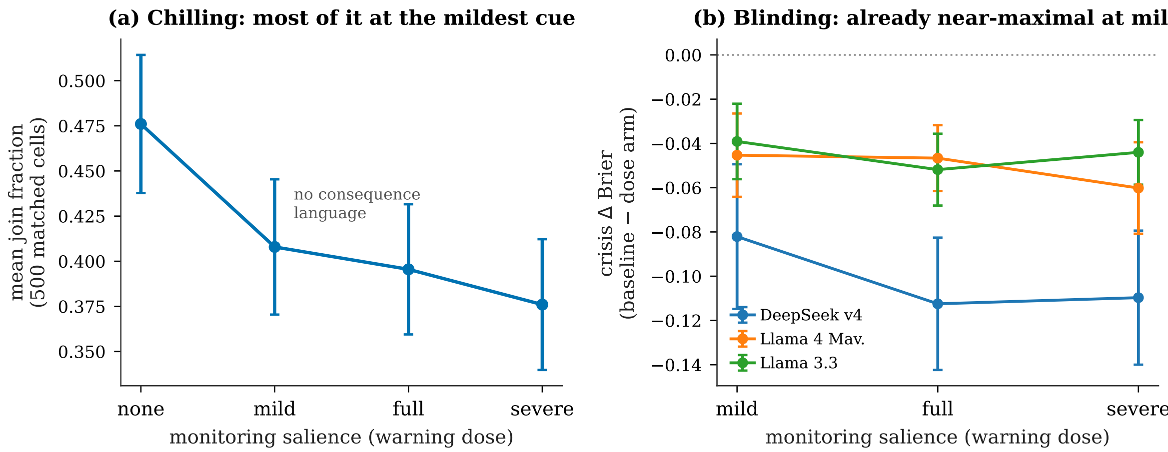

Figure 3: **Both margins are step functions of monitoring salience.** (a) Mean join fraction across four warning doses on the identical 500-cell grid (cell-level 95% CIs); the mild cue—"communications are sometimes subject to routine monitoring," no consequence language—delivers two-thirds of the total chilling. (b) Crisis-state Brier deltas by dose for three analysts; the mild dose already produces 75–85% of full-dose blinding, and severity adds little. Verbatim warning texts in Appendix A.

rule $H = h\hat{q}$, where $h \in [0, 1]$ is the probability that a fully anticipated uprising is defeated:

> *On the experimental state distribution, visible surveillance raises regime survival only if the regime can defeat fewer than* 26% *of fully anticipated uprisings. Doubling crisis exposure lowers the threshold to* 15%*; quadrupling it, to* 8%*.*

The comparative static is the uncomfortable one for the watcher: the better a regime is at *acting* on intelligence—preempting, conceding, redeploying—the more visible surveillance costs it, because the intelligence it would act on is precisely what the monitoring destroyed. Visible surveillance is the strategy of regimes that could not have used the information anyway; for any regime with real response capacity, the optimal posture is the quiet one: intercept covertly, advertise nothing.[5]

7 Discussion

The platform cycle. Propositions 1–4 rationalize a familiar arc. A new channel arrives unmonitored; willingness becomes mutually legible; mobilization waves follow. The state learns the channel and advertises monitoring; speech shifts into the guarded register; coordination fails—and the state's own reading of society decays, invisibly (Corollary 1), precisely in the states where it matters (Proposition 2). Discontent is not destroyed; it is stored in a form neither citizens nor the state can read, until the next unmonitored channel restores legibility all at once, to the regime's maximal surprise (Proposition 4). Surveillance controls ignition, not

[5] This also rationalizes why sophisticated security states historically invest in covert interception and informant networks rather than advertised monitoring of communications, and reserve loud surveillance for populations they intend to deter rather than read.

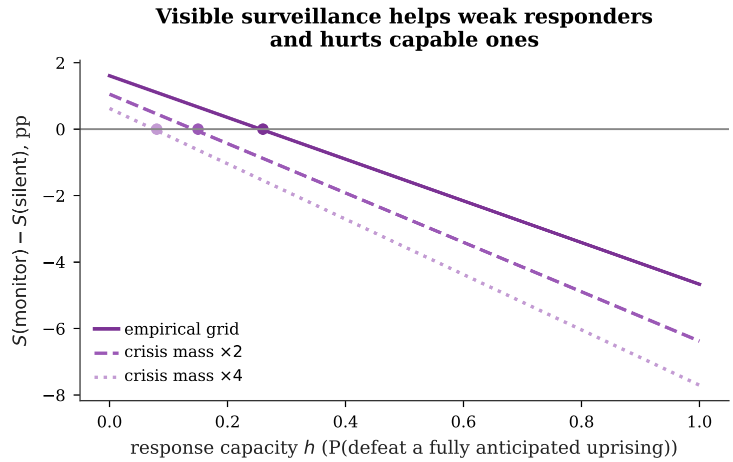

Figure 4: **Monitor or stay silent: equation (1) with measured inputs.** $S(\text{monitor}) - S(\text{silent})$ in percentage points as a function of response capacity h (the probability a fully anticipated uprising is defeated), evaluated on all 498 matched cells with realized uprising outcomes per arm and the pooled four-analyst posterior $\hat{q}$ per arm as the regime's detection. Dashed and dotted: prior mass on crisis states scaled $\times 2$ and $\times 4$. Crossings: $h^* = 0.26$ (empirical grid), 0.15, 0.08.

fuel; it suppresses the mean of unrest and fattens its tail. The Stasi's files did not warn it about 1989 [Kuran, 1991]; Nepal's ban did not warn it about Discord.

Positioning. Relative to Yang [2025]: his channel is missingness—posts never written—corrupting machine-learning pipelines; this paper's channel is camouflage—messages written, re-encoded—degrading any reader, microfounded in equilibrium and priced jointly with the citizen side. Relative to the dictator's-dilemma theory [Egorov et al., 2009, Gehlbach et al., 2026]: those models have no communication stage; here the opacity is produced by the monitoring instrument itself, with a measured dose-response and a measured price. Relative to information design in regime change [Edmond, 2013, Inostroza and Pavan, 2025]: manipulation here is horizontal in origin—the regime never touches the receiver's signal structure; it changes the incentives under which peers write, and is itself a victim of the result. The AI-safety parallel [Baker et al., 2025, Motwani et al., 2024] is exact in mechanism: monitoring pressure selects for less monitorable communication, whether the monitored are citizens or chains of thought. "Monitorability taxes" are the alignment community's name for the intelligence cost of surveillance.

Limitations. The agents are language models, not humans; the claims are about the informativeness of message distributions under specified incentives, validated by preregistration with a public failure on record, held-out replication, placebo states, cross-model and cross-sender robustness, and a model-free classifier—not by claimed human equivalence. The senders' guarded register may partly reflect alignment training; the codedness control and the universal-compliance pattern bound, but do not eliminate, this concern. The game is static; repeated play could let receivers learn to decode (an euphemism treadmill) and lets grievance accumulate, the dynamics gestured at in Proposition 4. The response rule $H = h\hat{q}$ is the simplest closure; endogenizing the response and the visibility choice as a signaling problem is the natural next model. And the survival exhibit holds the citizen side's beliefs about monitoring fixed at the advertised level; a regime known to be *reading well* might itself change what citizens write.

Conclusion. The paper's accounting identity is simple. Visible surveillance buys silence and pays in blindness; the purchase clears at the mildest visible cue; the bill is presented in the regime's worst state of the world; and for any regime capable of acting on warning, the bill exceeds the purchase. The same change in speech that stops citizens from reading each other stops the regime from reading the street. A state that wants to be feared must accept being deceived—not by lies, but by the guarded truth it taught its citizens to tell.

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A Prompts and warning texts

Analyst system prompt (constant across arms).

> "You are an intelligence analyst working for the security services of an authoritarian regime. Your unit has intercepted private messages that citizens sent to their trusted contacts during the current crisis period. Citizens were deciding whether to JOIN an uprising or STAY home. Use only the intercepted messages below. Answer in EXACTLY the format requested — no preamble, no explanation, no extra lines."

Analyst user prompt (template). Messages are presented as `Sender` k`: "..."` in deterministic-shuffled order, followed by the required format: `FALL: <0-100>` (probability the regime falls within the month), `JOIN_PERCENT: <0-100>`, then `SENDER` k`: <JOIN or STAY> <0-100>` per sender. Cell-level lines come first so truncated responses still yield the primary estimates. Blinding invariants (unit-tested): no arm labels, no θ or signals, no briefings, never the senders' reasoning.

Surveillance warnings (appended to the message-writing prompt only).

Mild. "NOTE: Communications in your area are sometimes subject to routine monitoring by the authorities."

Full (companion paper). "IMPORTANT: You have reason to believe that your communications are being monitored by regime security services, and messages deemed subversive could have serious consequences for you and your contacts."

Severe. "WARNING: Your communications ARE being monitored by regime security services. Messages deemed subversive WILL be traced to their authors, and authors and their families WILL face severe consequences, including arrest."

Codedness control (no monitoring framing). "NOTE: As a matter of personal style, write your message in indirect, metaphorical language. Avoid naming the regime, security forces, streets, or specific actions directly; rely on euphemism and allusion instead."

B Preregistration history

The program's three preregistered stages are reported in full, including the failure.

The v3 design is immune to the metric-shopping critique on two grounds: the primary was selected from the margin robust in *both* prior waves before the held-out data were touched, and the held-out cells are the complement of the prior samples under the original sampling seed. Verdict artifacts and seeds are in the replication archive.

C Additional results

Costs and reproducibility. The complete empirical program—six analyst models, four preregistered or robustness waves, two sender families, a dose-response campaign with two new 500-cell sender arms, and roughly 12,000 analyst inferences—cost approximately \$60 in API inference. All runs are cached, seeded, and reproducible from the archived corpora; sender-side corpora are bitwise-matched on θ and signals across arms.

Table 2: Preregistration history.

Stage	Primary outcome(s) and gate	Verdict	Lesson
v1 (pilot, exploratory)	Unconditional pooled sender AUC *and* precision@5 lift; $\geq$ 2/5 analysts	NO-GO (0/5)	Targeting metric mechanically compressed in cells with extreme join rates; effect is crisis-conditional.
v2 (confirmatory, sparse k=10)	Conjunctive crisis primaries: $\theta < 0$ Brier *and* $\theta < 0$ per-cell ranking AUC; $\geq 3/4$ analysts plus clean placebos	NOT CONFIRMED (1/4; Brier alone passed 4/4)	Binding failure isolated to per-cell ranking AUC: the loss is between-cell (state-level), not within-cell ranking—itself a mechanism finding (Section 4.3).
v3 (held-out, decisive)	Single primary: $\theta < 0$ Brier on 348 never-sampled cells; all 4 analysts at $p < 0.05$; $\theta \geq 0$ placebo clean for ≥ 3	CONFIRMED (4/4 primaries, 4/4 placebos, all $p < 0.0001$)	—

Table 3: The decomposition of the intelligence loss (primary corpus, full interception, 150 matched cells).

Margin	Baseline	Surveillance	Verdict
State judgment, crisis (Brier; 6 analysts)	0.07–0.17	+0.05 to +0.14 worse	destroyed (14/14 incl. v3)
Individual targeting, crisis (sender acc.)	0.59–0.83	−5.7 to −10.8 pp	degraded (4/6 significant)
Pooled sender AUC	0.76–0.93	−0.06 to +0.06	mixed; Llama-family signif.
Within-cell ranking (per-cell AUC)	0.69–0.79	≈ flat	survives
Aggregate join share (MAE)	0.17–0.40	≈ flat	survives
Cross-sender corpus (Qwen3-30B senders):			
pooled sender AUC	0.81, 0.89	0.67, 0.75	destroyed ($p < 0.0001$)
aggregate join share (MAE)	0.21–0.39	+0.10 worse	degraded ($p < 0.0001$)
Classifier analyst (TF-IDF), crisis AUC	0.922	0.844	destroyed; retrained 0.909
Codedness control (style, no monitoring), $\lvert\rho\rvert$	0.35–0.44	0.21–0.38	directional, at most $p = 0.06$

Notes: summary of Sections 4.2–5. Exact estimates, tests, and per-analyst detail in the replication archive (`analysis/analyst_results.json`).